Ge 158 Lecture 8: Scattering from the Ocean and Other Surfaces with Small-Scale Roughness

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Topics covered

- Ocean wave spectra
- Scattering from slightly rough surfaces (Bragg scattering)
- Scattering from undulating, slightly rough surfaces

1 Overview

- We will use scattering of EM waves from the ocean surface as an illustrative example of scattering from a broad range of natural surfaces.
- Advantages of doing so:
  - allows us to incorporate concepts from the beginning of the semester on dielectric properties (Lecture 1), because temperature and salinity have important effects on scattering from the ocean;
  - nicely illustrates the influence of surface roughness of varying wavelengths and sources.
- To get started, let us consider the geometry of the surface and the processes that influence it.

2 The ocean surface

2.1 What sets the elevation of the ocean surface

- The elevation of the ocean surface is influenced by a variety of factors. Roughly in order of decreasing wavelength:
  - the gravitational equipotential surface (geoid), which approximates mean sea level and is the elevation that the ocean would take given a planet's mass distribution and rotation, and in the absence of external forcing (tides, winds, etc.);
  - tides;
  - atmospheric pressure gradients;
  - currents;
  - ★ wind-driven waves → of greatest interest in today's lecture.

2.2 Wind-driven surface waves

- Ocean surface waves are mechanical waves that exist at the ocean–atmosphere boundary.
- As with any mechanical wave, they are defined by a perturbing force and a restoring force.
- In all cases of interest in this lecture, the perturbing forces are shear stresses caused by the wind.
- We will focus our attention on two types of waves defined by different restoring forces:
  1. Gravity waves are long wavelength (> 2 cm) whose restoring force is gravity.
     - Swells, which build up from wind forcing over long timescales, are examples of gravity waves.
     - Can play an important role in scattering from the ocean surface.
     - Terrestrial analogs: rolling hills, crop rows.
  2. Capillary waves are short wavelength (< 2 cm) whose restoring force is surface tension.
     - Produce small-scale roughness that is superimposed on longer-wavelength roughness.
     - Play a leading role in EM scatter from the ocean surface.
     - Terrestrial analog is any small-scale roughness: rocks, small erosional features.
- At radar wavelengths, it is often the case that the portion of the wave spectrum of greatest interest lies at intermediate wavelengths.
- At these intermediate wavelengths, both gravity and surface tension act to restore the ocean surface $\rightarrow$ such waves are called gravity–capillary waves.

2.3 Dispersion relation for gravity–capillary waves

- The dispersion relation for gravity–capillary waves in deep water (depth $\gg$ wavelength) is given as

$$\omega^2 = gK + \frac{\gamma_{\rm st}}{\rho_w} K^3, \tag{1}$$

  where

  - $g \equiv$ gravitational acceleration;
  - $\gamma_{\rm st} \equiv$ surface tension;
  - $\rho_w \equiv$ mass density of water;
  - $\omega \equiv$ angular frequency and $K \equiv$ wavenumber ($= 2\pi/\Lambda$) of the ocean-surface wave.
- In the limits
  - small K: $\omega \rightarrow \sqrt{gK}$ for gravity waves;
  - large K: $\omega \rightarrow \sqrt{\gamma_{\rm st} K^3/\rho_w}$ for capillary waves.
- The dispersion relation is what links the ocean-wave spectrum, usually measured as a function of frequency (e.g., Eq. 4 below), to the spatial roughness spectrum that EM scattering depends on (Lecture 7): wave energy in a band $d\omega$ belongs to the wavenumbers dK given by Eq. 1 (together with the spread of wave directions, this gives $W(k_x, k_y)$).

- We can see from this expression how "surfactants" (substances that reduce surface tension) change the dispersion relation of ocean surface waves ⇒ change the scattering properties of EM waves. Surface films also strongly damp the short waves, which reduces the roughness at the Bragg wavelength (Section 3) and makes slicks dark in radar images (Alpers and Hühnerfuss, 1989).

- Common examples of surfactants include mineral oil (as in the oil spill example that we have seen many times so far in this course) and biological films. Microplastics may also act as a surfactant, but I believe this remains an open question.

2.4 Phase velocity

- Gravity–capillary waves are excited (grow) in a turbulent wind field; the dominant (most energetic) waves reach equilibrium when the wind speed matches their phase velocity. From the dispersion relation, we define the phase velocity

$$u_p = \frac{\omega}{K} = \sqrt{\frac{g}{K} + \frac{\gamma_{\mathrm{st}}}{\rho_w} K}. \tag{2}$$

- In general, surfactants will reduce the surface tension of clean water ⇒ (all else being equal) will slow (gravity–)capillary waves (Figure 1).

- The centimeter-scale waves that dominate radar backscatter travel at only 0.2–0.6 m/s (Table 1), far slower than the wind. They grow and decay quickly, so they stay in equilibrium with the local wind, whereas long waves (swell) can arrive from distant storms. This is why radar backscatter measures the local wind.

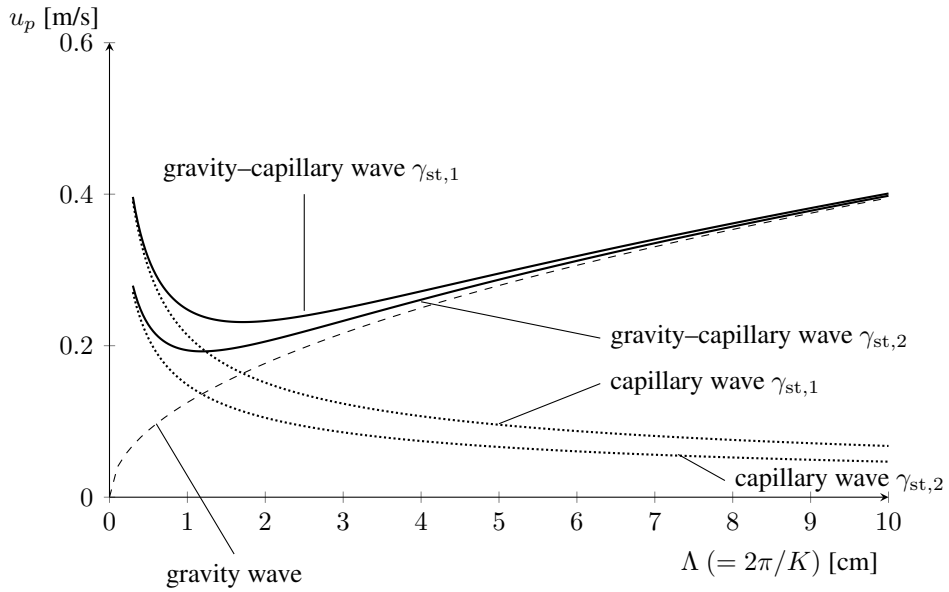


Figure 1: Phase velocity of deep-water gravity–capillary waves (solid, Eq. 2) versus wavelength for clean water ($\gamma_{\mathrm{st},1} = 0.0728$ N/m) and for water whose surface tension has been reduced by a surfactant ($\gamma_{\mathrm{st},2} = 0.035$ N/m, illustrative). Dashed: gravity-wave limit $\sqrt{g/K}$; dotted: capillary-wave limits $\sqrt{\gamma_{\mathrm{st}} K/\rho_w}$.

2.5 Wave height, the wave spectrum, and wind

- When wave speed (phase velocity) matches wind speed, the RMS height h of the ocean can be defined as

$$h^2 = \int_0^\infty \Psi(\omega)\, d\omega, \tag{3}$$

where the frequency spectrum Ψ is related to the local wind speed U (measured 19.5 m above the sea surface) through the empirical relation

$$\Psi(\omega) = \frac{8.1 \times 10^{-3}\, g^2}{\omega^5} \exp\left\{ -\frac{3}{4} \left(\frac{g}{U\omega} \right)^4 \right\} \tag{4}$$

$\Rightarrow$ strong correlation between wind speed and surface height, which, of course, we all know from experience. (Eq. 4 is the Pierson–Moskowitz spectrum for a fully developed sea.)

- Eq. 4 describes the energetic long waves: at $U = 10$ m/s, $h \approx 0.5$ m and the peak wavelength is ≈ 80 m. Its short-wave tail, converted to wavenumber with the dispersion relation, falls as K^{-3} (in the gravity range) and does not depend on U. The wind dependence that matters for radar is in the centimeter-scale waves, whose spectral level rises with wind stress (Ulaby and Long (2014), Fig. 16-6); this is the origin of the AU^n term below.

- For remote sensing, we are concerned with the direction of the winds relative to the look direction (or plane of incidence of the EM wave) of the sensor. This relation is usually given as an empirical relation between the normalized radar cross section σ^0_{pp} and the angle ϕ_{wind} between the plane of incidence and the wind direction:

$$\sigma^0_{pp} = AU^n \left(1 + a\cos\phi_{\text{wind}} + b\cos 2\phi_{\text{wind}}\right), \tag{5}$$

where A, a, b, n are all inferred from observations. This inference requires data collected from multiple look directions to provide info on wind speed and direction.

3 Simple models of EM scatter from the ocean (and other randomly rough surfaces)

- Let's dig a bit more into the mechanisms of EM scattering from the ocean surface.

- We'll start with the most useful model for surface scattering (at least at microwave frequencies): Bragg scattering or the small-perturbation model.

- Applicable when the RMS height of the surface and the horizontal length scale of surface roughness are small compared with the EM wavelength; canonically

$$kh < 0.3 \qquad \text{and} \qquad \frac{h}{\ell} < 0.3 \tag{6}$$

(Ulaby and Long (2014), p. 428, also require $k\ell < 3$).

- For the ocean, h here is not the rms height of Eq. 3 (about 0.5 m at $U = 10$ m/s, so $kh \gg 0.3$ at any microwave wavelength). In the Bragg model, h is the rms height of the small-scale roughness only, and h and W enter Eq. 9 only through $h^2 W(K_{\text{Bragg}}, 0)$, the amount of roughness at the component of the roughness spectrum responsible for the majority of scattering $\rightarrow$ called the Bragg wavelength. The long waves that carry most of the rms height are handled separately, as tilts (Section 4).

- Basic idea of the model: any random surface can be decomposed into a series of sinusoids. The wavelength that contributes most to the backscatter power is the one that maximizes constructive interference (resonance) of scattered waves (Figure 2).

3.1 Bragg scattering

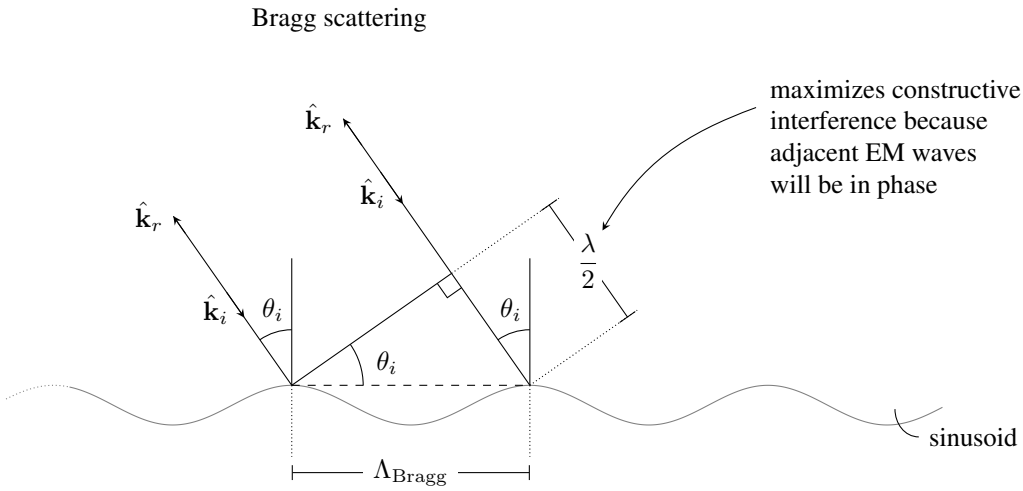


Figure 2: Geometry of Bragg scattering from a sinusoidal surface of wavelength Λ_{Bragg} for a wave incident at θ_i. The path to the farther crest is longer by $\Lambda_{\mathrm{Bragg}} \sin\theta_i = \lambda/2$ each way, so the backscattered waves from adjacent crests differ in path by one EM wavelength and add in phase.

- From geometry, we find

$$\Lambda_{\mathrm{Bragg}} = \frac{\lambda}{2\sin\theta_i} \tag{7}$$

or

$$K_{\mathrm{Bragg}} = 2k\sin\theta_i = \frac{2\pi}{\Lambda_{\mathrm{Bragg}}} \equiv \text{Bragg wavenumber.} \tag{8}$$

- At moderate incidence angles, the Bragg waves of microwave radars are centimeter- to decimeter-scale waves (Table 1): at Ku- to C-band they are gravity–capillary waves, in which both gravity and surface tension matter, and at L-band they are short gravity waves. This is why the dispersion relation, surface tension, and surfactants matter for radar.

- Because the Bragg waves move, the backscattered signal is Doppler shifted by $\pm 2u_p \sin\theta_i/\lambda$ (waves approaching and receding from the radar, with u_p from Eq. 2 at K_{Bragg}). Using Eq. 8, this is $\pm\omega/2\pi$, the frequency of the Bragg ocean wave from Eq. 1 (Table 1). (This is the two-way Doppler shift, $2/\lambda$ times the speed along the line of sight, $u_p \sin\theta_i$; Lecture 14.) Doppler spectra of sea echo at HF show these two sharp "Bragg lines" (Crombie, 1955), direct evidence that the radar sees the Bragg waves; at microwave frequencies the lines are broadened by the orbital motion of the longer waves. The offset of the lines from $\pm\omega/2\pi$ is used to measure surface currents.

3.2 The first-order small-perturbation model

- To apply the concept of Bragg scattering to backscattered power, Rice (1951) applied perturbation methods and derived an expression for the first-order (meaning, higher-order terms in the Taylor series expansion are dropped) normalized radar cross section

$$\sigma^0_{pp} = 8k^4h^2\cos^4\theta_i\,|\alpha_{pp}|^2\,W\left(K_{\mathrm{Bragg}}, 0\right), \tag{9}$$

Table 1: Bragg waves at common radar bands for $\theta_i = 35^\circ$ (Eqs. 1, 2, and 7; clean water). The ratio compares the gravity and surface-tension terms of Eq. 1; it is 1 at $\Lambda = 1.7$ cm.

Band	λ [cm]	Λ_{Bragg} [cm]	$gK \big/ \left(\gamma_{\text{st}} K^3/\rho_w\right)$	u_p [m/s]	$\omega/2\pi$ [Hz]
Ku (13.6 GHz)	2.2	1.9	1.3	0.23	12
X (9.6 GHz)	3.1	2.7	2.5	0.24	8.9
C (5.3 GHz)	5.7	4.9	8.3	0.29	6.0
L (1.26 GHz)	23.8	20.7	147	0.57	2.8

where $W(k_x, k_y) \equiv$ surface roughness spectrum discussed in the previous lecture

$$\left(W = \frac{\ell^2}{2} \exp\left\{ -\left(\ell k \sin\theta_i\right)^2 \right\} \quad \text{for a Gaussian surface} \right). \tag{10}$$

- The scattering coefficients are

$$\alpha_{hh} = \rho_h \equiv \text{Fresnel reflection coefficient for } h\text{-polarized waves at oblique incidence}, \tag{11}$$

$$\begin{aligned} \alpha_{vv} &= \rho_v - \tau_v^2 \frac{(\varepsilon_2 - 1)\tan^2\theta_i}{2} \\ &= (\varepsilon_2 - 1) \frac{\sin^2\theta_i - \varepsilon_2\left(1 + \sin^2\theta_i\right)}{\left(\varepsilon_2 \cos\theta_i + \sqrt{\varepsilon_2 - \sin^2\theta_i}\right)^2}, \end{aligned} \tag{12}$$

where ρ_v and τ_v are the Fresnel reflection and transmission coefficients for v-pol at oblique incidence (Lecture 5), and we have assumed $\varepsilon_1 \approx 1$.

3.2.1 Why does $\alpha_{hh} = \rho_h$ but $\alpha_{vv} \neq \rho_v$?

A. Go back to EM boundary conditions: recall from Faraday's law we got the condition that the tangential component of the electric field is continuous across the surface (Lecture 4); the h field is entirely tangent to the surface for all roughness (at least to a reasonable approximation), whereas the tangential component of the v field changes with roughness.

3.2.2 Key take-aways

1. $\sigma^0_{hv} = \sigma^0_{vh} = 0$ to first order (higher-order factors introduce small nonzero values of σ^0_{hv} (and σ^0_{vh}), but these are generally negligible as they are in the noise of RS instruments).

2. α_{pp} represents total scattered power and is

   (a) polarization-dependent;

   (b) a function of dielectric properties and viewing geometry (indirectly EM wavelength);

   (c) independent of characteristics of surface roughness.

3. W influences the direction of scattering and is

   (a) dependent on characteristics of roughness;

   (b) dependent on EM wavelength and viewing geometry;

(c) independent of dielectric properties;

(d) independent of polarization.

4. $\sigma^0_{vv} > \sigma^0_{hh}$ for $\theta_i > 0$ (at normal incidence $\alpha_{vv} = \alpha_{hh}$) due to the expression for reflectivity (thus, due to EM BCs).

- Back to the wind model, Eq. 5: the $\cos 2\phi_{\text{wind}}$ term arises because the short waves are anisotropic (they run mostly along the wind), so $W(K_{\text{Bragg}}, 0)$ depends on the look direction relative to the wind. The $\cos\phi_{\text{wind}}$ (upwind–downwind) term cannot come from first-order Bragg scattering, because $W(-k_x, -k_y) = W(k_x, k_y)$ for any real, stationary surface, whose correlation function is centrosymmetric (Ulaby and Long (2014), p. 789); it comes from effects beyond first-order Bragg scattering, such as the skewed shape of wind-driven waves (Ulaby and Long (2014), p. 790).

4 Tilted Bragg scatter model

- The previous model only applies to horizontal surfaces with small-scale roughness.
- We can combine the Bragg model with the Kirchhoff model for undulating surfaces (Lecture 7) by (again) treating the surface as a bunch of randomly rough surfaces (infinite extent) that are tilted due to longer-wavelength roughness features.
- On the ocean, this simply means that we represent the wind-driven gravity–capillary waves with a Bragg scatter model and rotate our local coordinate system to account for the passage of gravity waves (swells).
- This is a two-scale view of the wave spectrum of Section 2: waves much longer than Λ_{Bragg} (swell and the peak of Eq. 4, which carry most of h in Eq. 3) tilt the patches, while waves near Λ_{Bragg} set W. Wind speed enters through both.
- This is a very accurate, simple model of scattering from the ocean surface for incidence angles ranging from 30°–65°. (Lower incidence angles produce primarily specular scatter, while angles $> 65°$ involve different mechanisms that we won't discuss in this course because they are esoteric.)

4.1 Normalized radar cross section of a tilted patch

- If we define the tilting of the rough surface patches in terms of two angles
  - $\psi \equiv$ in-plane (plane of incidence) tilt,
  - $\zeta \equiv$ anti-plane (plane of incidence) tilt,

  it can be shown (Valenzuela, 1978) that the normalized radar cross section is

$$\sigma^0_{pq} = 8k^4h^2\cos^4\theta'_i\,|S_{pq}|^2\,W, \tag{13}$$

  where

$$W = W\big(2k\sin(\theta_i+\psi),\ 2k\cos(\theta_i+\psi)\sin\zeta\big), \tag{14}$$

  and the polarization-dependent reflectivities, written as squared magnitudes of the rotated scattering coefficients S_{pq}, are

$$|S_{pp}|^2 = \left|\left(\frac{\sin(\theta_i+\psi)\cos\zeta}{\sin\theta'_i}\right)^2\alpha_{pp} + \left(\frac{\sin\zeta}{\sin\theta'_i}\right)^2\alpha_{qq}\right|^2 \qquad \text{for } p \neq q, \tag{15}$$

$$|S_{hv}|^2 = |S_{vh}|^2 = \left(\frac{\sin(\theta_i + \psi) \sin\zeta \cos\zeta}{\sin^2 \theta_i'} \right)^2 |\alpha_{vv} - \alpha_{hh}|^2 \,, \tag{16}$$

with

$$\theta_i' = \cos^{-1}\left(\cos(\theta_i + \psi) \cos\zeta \right) \equiv \text{incidence angle measured relative to the surface normal,} \tag{17}$$

and α_{hh} and α_{vv} as defined in the Bragg model (Eqs. 11 and 12), evaluated at θ_i'.

4.2 Take-aways

- This model reduces to the previous (untilted) model when $\psi = \zeta = 0$.
- Take-aways:
  - Tilting modifies both reflectivity and the effective roughness spectrum.
  - But the dependencies of reflectivity and roughness spectrum remain the same as in the Bragg model and also include tilt angles.
  - Cross-polarized scattering (σ^0_{hv}) is nonzero when the anti-plane tilt angle $|\zeta| > 0$ (and $\theta_i + \psi \neq 0$)
    - $\rightarrow$ because in this case, neither the h-polarized nor v-polarized electric fields are locally tangent to the surface $\Rightarrow$ conservation of the tangential component of the $\mathbf{E}$ field results in a change of polarization.
- And those are the basic concepts of EM scattering from surfaces.

References

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Valenzuela, G. R. (1978). Theories for the interaction of electromagnetic and oceanic waves—A review. *Boundary-Layer Meteorology*, 13(1–4), 61–85. doi:10.1007/BF00913863.

Variables

Units are SI. Values are given for universal constants; "typical" values are representative, not definitive.

Symbol	Description	Units	Value
Ocean surface waves			
g	gravitational acceleration	m/s^2	9.81
γ_{st}	surface tension	N/m	0.0728 (clean water, 20°C)
$\gamma_{\mathrm{st},1},\ \gamma_{\mathrm{st},2}$	surface tension of clean water and of surfactant-covered water (Figure 1)	N/m	0.0728; 0.035 (illustrative)
ρ_w	mass density of water	kg/m^3	≈ 1000 (seawater ≈ 1025)
ω	angular frequency of an ocean-surface wave	rad/s	–
K	wavenumber of an ocean-surface wave	rad/m	–
Λ	wavelength of an ocean-surface wave, $2\pi/K$	m	gravity: > 2 cm; capillary: < 2 cm
u_p	phase velocity, ω/K	m/s	minimum 0.23 (clean water)
h	RMS height of the ocean surface (Eq. 3; Sec. 3: of the small-scale roughness only)	m	≈ 0.5 m at $U = 10$ m/s (Eq. 3)
$\Psi(\omega)$	frequency spectrum of the surface height	m^2 s	Eq. (4)
U	wind speed (19.5 m above the surface in Eq. (4))	m/s	–
ϕ_{wind}	angle between the plane of incidence and the wind direction	rad	–
$A,\ a,\ b,\ n$	empirical coefficients in Eq. (5)	varies	inferred from observations
Bragg scattering			
k	EM angular wavenumber in the medium above the surface	rad/m	$2\pi/\lambda$
λ	EM wavelength	m	–
θ_i	angle of incidence	rad	–
$\hat{\mathbf{k}}_i,\ \hat{\mathbf{k}}_r$	propagation unit vectors of the incident and backscattered waves	dimensionless	–
Λ_{Bragg}	Bragg wavelength of the surface roughness	m	$\lambda/(2\sin\theta_i)$
K_{Bragg}	Bragg wavenumber	rad/m	$2k\sin\theta_i$
ℓ	correlation length	m	–
σ^0_{pq}	normalized radar cross section, receive/transmit polarizations p, q	dimensionless	$\sigma^0_{hv} = 0$ (first order, untilted)
$W(k_x, k_y)$	roughness spectrum (Lecture 7)	m^2	–
$k_x,\ k_y$	spectral wavenumbers of the roughness spectrum (Lecture 7)	rad/m	–
$p,\ q$	receive and transmit polarization labels (h or v)	–	–
$\alpha_{hh},\ \alpha_{vv}$	first-order scattering coefficients	dimensionless	$\alpha_{hh} = \rho_h$
$\rho_h,\ \rho_v,\ \tau_v$	Fresnel reflection coefficients (h, v) and transmission coefficient (v)	dimensionless	–

Symbol	Description	Units	Value
$\varepsilon_1,\ \varepsilon_2$	dielectric constants above and below the surface	dimensionless	$\varepsilon_1 \approx 1$
Tilted Bragg model			
ψ	in-plane tilt of a rough patch (in the plane of incidence)	rad	–
ζ	anti-plane tilt of a rough patch	rad	–
θ_i'	local angle of incidence (from the patch normal)	rad	$\cos^{-1}[\cos(\theta_i + \psi)\cos\zeta]$
S_{pq}	rotated scattering coefficient of a tilted patch ($\lvert S_{pq}\rvert^2$: polarization-dependent reflectivity)	dimensionless	$S_{pp} = \alpha_{pp}$ if $\psi = \zeta = 0$